

Lecture 1 - 01/06/2025

Review:

Markov Process:

→ AR(1)

$$\hookrightarrow \underline{z_{t+1}} = \underline{\rho z_t} + \underline{\varepsilon_{t+1}} \quad \text{s.t.} \quad \varepsilon_{t+1} \sim N(0, \sigma_\varepsilon^2)$$

↳ limited memory process

Markov Chain:

→ Markov process w/ finite no. of values

$$\hookrightarrow z \in \{z_1, z_2, \dots, z_n\}$$

→ Obj. 1:

↳ set of states: $\{z_1, \dots, z_n\}$

→ Obj 2:

↳ transition matrix / probs b/w states

$$\hookrightarrow i, j \in \{1, \dots, n\} \Rightarrow p_{ij} = \text{prob}(z_{t+1} = z_j \mid z_t = z_i)$$

→ Obj 3:

↳ unconditional dist $\underline{\pi}_t$

↳ $\underline{\pi}$ can change over time

$$\hookrightarrow \underline{\pi}_{t+1}' = \underline{\pi}_t' \cdot P$$

↳

$$(\pi_{1t+1}, \pi_{2t+1}, \dots, \pi_{nt+1}) = (\pi_{1t}, \pi_{2t}, \dots, \pi_{nt}) \begin{pmatrix} p_{11} & \dots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \dots & p_{nn} \end{pmatrix}$$

↳ P is probability unconditional on today that state is reached tomo.

$$\hookrightarrow \pi_{1t+1} = \pi_{1t} \cdot p_{11} + \pi_{2t} \cdot p_{21} + \dots + \pi_{nt} \cdot p_{n1}$$

Stationary Invariant Distribution:

$$\rightarrow \pi_{t+1} = \pi_t = \pi^* \text{ s.t. } p_{ij} > 0 \quad \forall i, j$$

↳ can find via guess to converge to SID

↳ in MatLab: raise p to large power

Law of Iterated Expectations:

$$\rightarrow \exists y \Rightarrow \text{find } E[E(y_{t+2} | x_{t+1}) | x_t]$$

$$\hookrightarrow = E(y_{t+2} | x_t) \Rightarrow E_t[E_{t+1}(y_{t+2})] = E_t y_{t+2}$$

↳ only know t thus conditional for $t+2$
by conciseness

Geometric Series:

$$\rightarrow a + ar + ar^2 + \dots + ar^{n-1} = \sum_{k=0}^{n-1} ar^k = a \frac{1-r^n}{1-r} \quad (\text{finite})$$

$$\rightarrow \sum_{k=0}^{\infty} ar^k = a \frac{1}{1-r} \text{ s.t. } r < 1 \quad (\text{inf.})$$

Normal Dist:

$$\rightarrow f(x | \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

↳ AR(1)

$$\hookrightarrow z_t = \mu + \rho z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

↳ conditional Exp:

$$\hookrightarrow E_{t-1} z_t = E(z_t | z_{t-1}) = E_{t-1}(\mu + \rho z_{t-1} + \varepsilon_t) = \mu + \rho z_{t-1}$$

$$\hookrightarrow (E_{t-1}, \varepsilon_t) = 0$$

Normal Dist.



→ Conditional Variance:

$$\text{var}_{t-1}(z_t) = \text{var}_{t-1}(\mu_t + \rho z_{t-1} + \varepsilon_t) = \sigma_\varepsilon^2$$

→ unconditional expectation:

$$E(z_t) = E(\mu + \rho z_{t-1} + \varepsilon_t) = E\mu + \rho \overset{\text{rho}}{E(z_{t-1})} + E\varepsilon_t$$

$$\hookrightarrow E(z_t) = E(z_{t-1}) = E(z) \Rightarrow \text{expect it's stationary}$$

$$\hookrightarrow E(z) = \mu + \rho E(z) = \mu / (1 - \rho)$$

→ substitution

$$z_t = \mu + \rho z_{t-1} + \varepsilon_t$$

$$z_{t-1} = \mu + \rho z_{t-2} + \varepsilon_{t-1}$$

$$z_t = \mu + \rho\mu + \rho^2 z_{t-2} + \rho\varepsilon_{t-1} + \varepsilon_t$$

substitute until initial z_0

$$z_t = \mu + \rho\mu + \rho^2\mu + \dots + \rho^t z_0 + \varepsilon_t + \rho\varepsilon_{t-1} + \rho^2\varepsilon_{t-2}$$

$$z_0 = 0 \Rightarrow E(z) = \mu(1 + \rho + \rho^2 + \dots) = \frac{\mu}{1 - \rho} \quad \left. \begin{array}{l} \text{uncondition} \\ \text{var + Exp} \end{array} \right\}$$

$$\text{var}(z) = \sigma_\varepsilon^2 (1 + \rho^2 + \rho^4 + \dots) = \sigma_\varepsilon^2 \frac{1}{1 - \rho^2}$$

→ Conditional Dist of z_t :

$$z_{t+1}|z_t \sim N(\mu + \rho z_t, \sigma_\varepsilon^2) \quad - \left(\frac{z_{t+1} - \mu - \rho z_t}{\sigma_\varepsilon} \right)^2$$

$$f(z_{t+1}|z_t) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon^2}} \cdot e$$

Normal Dist:

→ unconditional dist of z_t :

$$f(z_t) = \frac{1}{\sqrt{2\pi} \frac{\sigma_z}{1-\rho^2}} \cdot e^{-\frac{(z_t - \frac{\lambda}{1-\rho})^2}{2 \left(\frac{\sigma_z^2}{1-\rho^2} \right)}}$$



Log-Normal Dist:

→ $X = e^{\mu + \sigma z}$, $z \sim N(0, 1)$

↳ X has log-normal

↳ $\log(x) \sim N(\mu, \sigma^2)$

↳ expectation:

$$E(X) = e^{\mu + \frac{1}{2}\sigma^2} \text{ (func. of mean and var)}$$

$$E(X^t) = e^{t\mu + \frac{1}{2}t^2\sigma^2}$$

↳ variance:

$$\begin{aligned} \text{var}(X) &= E(X^2) - E(X)^2 \\ &= e^{2\mu + \frac{1}{2}4\sigma^2} - e^{2(\mu + \frac{1}{2}\sigma^2)} = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1) \end{aligned}$$

$$\begin{aligned} \text{var}(X^t) &= E(X^{2t}) - E(X^t)^2 \\ &= e^{2t\mu + 2t^2\sigma^2} - e^{2(t\mu + \frac{1}{2}t^2\sigma^2)} = e^{2t\mu + t^2\sigma^2} (e^{t^2\sigma^2} - 1) \end{aligned}$$

↳ Mode $X = \mu$

Preview:

↳ consumption & saving

↳ partial eq: all given & how agents respond

↳ HW1 due date tbd